

Iron / Microcytic — iron studies & IRP/IRE regulation



1. IRP-1 / IRP-2 readers

WHAT YOU SEE

Two scholars 'IRP-1' and 'IRP-2' reading by IRE iron-response elements

WHAT IT ENCODES

Iron regulatory proteins (IRP-1, IRP-2) bind iron-response elements (IREs) on mRNA to set the cell's iron-handling proteins post-transcriptionally based on intracellular iron. When cell iron is LOW, IRPs are active and bind IREs; when iron is HIGH, IRPs disengage. This is CELLULAR (intracellular) regulation, distinct from systemic hepcidin control.

BOARD TRAP

WRONG: conflating the IRP/IRE system with hepcidin. IRP/IRE is intracellular and post-transcriptional (it tunes ferritin and transferrin-receptor synthesis WITHIN a cell); hepcidin is the SYSTEMIC hormone controlling ferroportin and whole-body iron. Boards test that these are two separate layers of control.

2. LOW iron page

WHAT YOU SEE

Open book LEFT page 'LOW': TfR1 up, ferritin down (down-arrows on ferritin/ALAS2)

WHAT IT ENCODES

LOW cellular iron → active IRP binds IREs → STABILIZES transferrin-receptor (TfR1) mRNA (more receptor → pull iron IN) and BLOCKS translation of ferritin mRNA (less storage). Net effect: the cell maximizes iron uptake and minimizes sequestration.

BOARD TRAP

WRONG: thinking ferritin rises when cellular iron is low. When iron is scarce the IRP blocks ferritin translation, so ferritin FALLS and transferrin receptor RISES — the opposite of storage mode. (IRE position matters: 5' IRE on ferritin = translation block; 3' IREs on TfR1 = mRNA stabilization.)

3. HIGH iron page

WHAT YOU SEE

RIGHT page 'HIGH': TfR1 down, FERRITIN up, FPN up, ALAS2 up

WHAT IT ENCODES

HIGH cellular iron → IRPs RELEASE the IREs → ferritin and ferroportin mRNAs are translated (store and export iron) while TfR1 mRNA is degraded (stop importing). ALAS2 (erythroid heme-synthesis enzyme) is also translated, channeling iron into heme.

BOARD TRAP

WRONG: expecting transferrin receptor to stay high when the cell is iron-replete. With abundant iron, TfR1 is DOWN (no need to import) while ferritin/ferroportin go UP — the inverse of the low-iron state. Mixing up the direction of these arrows is a classic exam error.

4. Hepcidin (PHAGY/Ft cart)

WHAT YOU SEE

Monster 'PHAGY' guarding a ferritin cart, 'LAST RESORT'

WHAT IT ENCODES

Macrophages recycle iron from senescent RBCs and either store it as ferritin or export it via ferroportin. Hepcidin is the systemic gatekeeper: HIGH hepcidin degrades ferroportin so macrophages HOARD iron (ferritin rises, serum iron falls) — the engine of anemia of chronic disease.

BOARD TRAP

WRONG: assuming a high ferritin always means iron overload that needs treatment. Macrophage iron HOARDING in inflammation also raises serum ferritin — distinguish true overload (high TSAT, hemochromatosis) from inflammatory ferritin elevation (low TSAT, low TIBC) before acting.

5. CANCER / ACD alcove (red)

WHAT YOU SEE

RED alcove: crab (cancer) spilling, 'CANCER'

WHAT IT ENCODES

Inflammation and malignancy raise IL-6 → hepcidin → iron trapped in macrophages. Iron studies in ACD: serum iron LOW, TIBC LOW, ferritin HIGH/normal, TSAT low, sTfR normal. Usually normocytic, can be mildly microcytic. The cure is treating the underlying disease.

BOARD TRAP

WRONG: expecting a LOW ferritin in anemia of chronic disease. Ferritin is HIGH/normal in ACD because iron is sequestered and ferritin is an acute-phase reactant; a low ferritin would instead indicate true iron deficiency. Low TIBC is the other tip-off separating ACD from IDA.

6. ATHLETE alcove (yellow)

WHAT YOU SEE

YELLOW alcove: runner, 'ATHLETE', 'MILD'

WHAT IT ENCODES

Endurance athletes can show a MILD apparent anemia from two mechanisms: plasma-volume expansion causing dilutional 'sports anemia' (a false low Hb with normal red-cell mass), AND true iron loss (foot-strike hemolysis, GI microbleeding, sweat losses, hematuria) plus exercise-induced hepcidin spikes.

BOARD TRAP

WRONG: treating dilutional pseudo-anemia with iron. If iron studies are normal, the low Hb is a benign hemodilution that needs no iron; only treat when ferritin/TSAT confirm true deficiency. Conversely, don't dismiss a fatigued athlete — check ferritin, since real iron loss is common.

7. MASKED IDA (purple)

WHAT YOU SEE

PURPLE alcove: masked figure, 'MASKED', high-ferritin disguise

WHAT IT ENCODES

Concurrent inflammation can falsely NORMALIZE or RAISE ferritin and MASK coexisting iron deficiency. To unmask it, use markers unaffected by inflammation: soluble transferrin receptor (sTfR), the sTfR/log-ferritin index, reticulocyte Hb (ChR), or a bone-marrow iron stain. In inflammation, a ferritin up to ~70–100 may still mean depleted stores.

BOARD TRAP

WRONG: ruling out iron deficiency because ferritin is 'normal' (e.g., 80) in a sick/inflamed patient. The inflammation-adjusted cutoff is higher, and ferritin can be falsely reassuring — order sTfR or marrow iron when the clinical picture still suggests deficiency.

8. RACE / baseline (blue)

WHAT YOU SEE

BLUE alcove: two elderly faces, 'RACE', '-1'

WHAT IT ENCODES

The reference Hb threshold for 'anemia' varies by sex, age, and ancestry (WHO: <13 g/dL men, <12 g/dL non-pregnant women, <11 in pregnancy). Individuals of African ancestry may run slightly lower baseline Hb. Knowing a patient's own baseline matters as much as the population cutoff.

BOARD TRAP

WRONG: applying a single Hb cutoff to everyone. A 'normal' value may be anemic for one patient (a drop from their baseline) and normal for another; conversely a low-normal value in a person with a constitutionally lower baseline may not be pathologic. Trend against the patient's own prior values.

9. Insect-bite / hookworm

WHAT YOU SEE

Prevention panel 1: mosquito/insect with warning triangle

WHAT IT ENCODES

Hookworm (*Ancylostoma duodenale*, *Necator americanus*) and other GI parasites cause chronic intestinal blood loss and are the leading cause of IDA in the developing world. Look for travel/endemic exposure, eosinophilia, and stool ova/parasites; treat with albendazole/mebendazole plus iron.

BOARD TRAP

WRONG: omitting parasite/exposure history in IDA from an endemic region or recent traveler. Hookworm IDA is treated by ERADICATING the parasite plus iron repletion — giving iron alone without deworming lets blood loss continue and the anemia recur.

10. Transfusion >200 / iron overload

WHAT YOU SEE

Panel 2: blood bag '>200', calendar

WHAT IT ENCODES

Each unit of RBCs delivers ~200–250 mg iron and the body cannot excrete it, so chronic transfusion (and ineffective erythropoiesis in thalassemia/sideroblastic/MDS) causes iron OVERLOAD. Manage with iron chelation (deferasirox, deferoxamine, deferiprone), monitoring ferritin and cardiac/hepatic iron (MRI T2*).

BOARD TRAP

WRONG: giving iron to a chronically transfused or thalassemia-major patient. They are iron-OVERLOADED, not deficient — added iron accelerates cardiac/hepatic damage. These patients need CHELATION, not supplementation; never reflexively replete iron based on anemia alone.

11. Screen first / 6 wk

WHAT YOU SEE

Panel 3: pencil, '6 WK', 'SCREEN FIRST', stool/colon kit

WHAT IT ENCODES

In men and postmenopausal women, newly found IDA mandates a search for the SOURCE first — bidirectional endoscopy (EGD + colonoscopy) for occult GI blood loss/malignancy, since right-sided colon cancer often presents purely as IDA. Repleting iron is secondary to localizing the bleed.

BOARD TRAP

WRONG: treating IDA and stopping there without an evaluation for the cause. Correcting the anemia while leaving an undiagnosed colon cancer or ulcer to progress is the classic board pitfall — in these populations, IDA is a colon cancer screen until proven otherwise.

12. Pregnancy supplementation

WHAT YOU SEE

Pregnant figure with supplement bottle/checklist

WHAT IT ENCODES

Pregnancy roughly triples iron demand (fetus, placenta, expanded maternal red-cell mass; ~27 mg/day recommended). Routine screening and prophylactic iron are standard; untreated maternal IDA risks preterm birth, low birth weight, and impaired infant neurodevelopment.

BOARD TRAP

WRONG: interpreting the mild physiologic Hb dip of pregnancy as no concern, or conversely treating it as severe disease. Pregnancy causes dilutional pseudo-anemia from plasma-volume expansion, but TRUE iron deficiency is also very common — confirm with ferritin and supplement rather than assuming it is purely dilutional.